# HVAC Troubleshooter

Starter Troubleshooting Materials — Field-Derived Drafts

Purpose: Build a searchable offline reference for commercial hydronic HVAC troubleshooting. These materials are based on previously discussed field cases and are preliminary diagnostic aids, not manufacturer-approved procedures. Verify all fault meanings, limits, tests, and repair steps against the exact equipment manual and site requirements before field use.

## 1. Recommended Modules

### Tekmar controls and outdoor reset

- Tekmar 256: outdoor temperature readings changed from approximately 65°F to 82–85°F.
- Tekmar 274: four-boiler system displayed approximately 62°F when actual outdoor temperature was about 54°F.
- Tekmar 671: outdoor sensor open-circuit alarm, Internet weather-source configuration, missing time, and weather data not displaying.
- Investigate sensor source, sensor wiring and resistance chart, network/weather configuration, WWSD, and approved boiler test behavior.

### B&G; Ecocirc pump faults

- E02: high motor current, repeated fault after power cycling, and possible pump/motor condition concerns.
- E06: reported incoming voltage imbalance outside the pump’s allowable range; investigate supply conditions, connections, and generator operation.
- Record the exact model and fault before resetting; a successful restart does not prove the underlying issue is resolved.

### Danfoss VFD commissioning and programming

- Confirm exact VLT drive model and whether the reported identifier is a parameter number, display fault, or external-document reference.
- Verify motor nameplate data and sensorless commissioning data against the applicable procedure.
- Confirm pressure setpoint and system requirements before changing pressure-related settings.
- Record parameter changes and verify startup and operation using the approved commissioning process.

### AERCO BME boiler controls

- External interlock alarm despite apparent electrical continuity.
- Controller parameter backup, reset/restoration, and post-change verification.
- Distinguish programming/configuration concerns from wiring, interlock, and BMS issues.
- Never bypass an interlock or safety device to force operation.

## 2. Decision Tree A — Tekmar Outdoor Temperature Incorrect or Unstable

### Start: displayed outdoor temperature is incorrect

- 1. Identify the outdoor-temperature source: physical sensor, network, or Internet/weather service as supported by the exact controller.
- 2. Compare the displayed value with an independent outdoor temperature measurement and record the exact model and settings.
- 3. If stable but incorrect, verify the selected source, sensor specification, resistance-versus-temperature chart, and sensor placement.
- 4. If the value jumps, inspect connections, moisture, shorts, intermittent opens, and possible electrical interference.
- 5. If missing or invalid, verify input selection and network/weather-source configuration.
- 6. If testing a physical sensor, isolate the circuit as directed by the manual and compare measured resistance at the measured sensor temperature with the correct manufacturer chart. Do not assume all Tekmar sensors use the same resistance curve.
- 7. Verify recovery: stable, plausible reading; review reset curve and WWSD settings; verify boiler operation using the approved test procedure.

### Related field observation

- Tekmar 256 case, October 5, 2026: outdoor reading reportedly changed from about 65°F to 82–85°F. UP-button behavior was also reported; verify intended test-mode behavior in the exact manual.
- Tekmar 274 case, October 7, 2026: display reportedly showed 62°F while actual outdoor temperature was about 54°F. A measured sensor resistance was reported, but its interpretation requires the exact sensor chart and measurement conditions.
- Tekmar 671 case, October 9, 2026: outdoor sensor open-circuit alarm was present while the outdoor source was set to Internet. Time was missing until set to AUTO; the alarm cleared and an update began. Record this as an observed sequence, not proof of root cause.

## 3. Decision Tree B — B&G; Ecocirc E02 / E06

### Start: pump stops or requires manual reset

- 1. Before resetting, record exact model, full code, operating conditions, supply measurements, and whether the fault occurs at startup or during normal operation.
- E02 — high motor current (field-reported meaning): isolate power and inspect for mechanical obstruction or binding only as directed by the service manual. Check pump/motor condition and supply conditions. If the fault persists, follow manufacturer service and replacement criteria.
- E06 — voltage imbalance (field-reported meaning): using qualified electrical procedures, measure applicable incoming supply conductors and inspect connections and source conditions. Compare readings with the exact model's allowable limits. Investigate generator operation if the fault correlates with generator use.
- 2. Correct the identified issue, clear the fault only as instructed, and observe startup and normal operation.
- 3. Do not treat a successful power cycle as proof the underlying issue is resolved.

### Documentation note

- Exact fault descriptions, reset instructions, electrical limits, and replacement criteria must be cited from the applicable Ecocirc manual before these entries are promoted to verified manufacturer guidance.

## 4. Decision Tree C — AERCO BME External Interlock Alarm

### Start: boiler reports external interlock alarm

- 1. Record exact alarm text, boiler designation, controller revision, operating state, and active BMS commands. Determine whether the alarm is active, latched, or historical according to the manual.
- 2. Verify the interlock circuit using the unit-specific wiring diagram and approved electrical procedures. Continuity alone may not prove correct operation under operating conditions.
- 3. Inspect field wiring, terminals, external contacts, BMS logic, and the prescribed signal/circuit state. Compare controller configuration with the documented installation.
- 4. If configuration changes are justified, back up existing parameters first and follow AERCO's authorized reset/restoration procedure.
- 5. Verify that the interlock operates correctly, the alarm clears by the prescribed method, and the boiler runs normally with all safety circuits intact.
- Safety: never bypass the interlock to force operation.

### Related field observation

- BME boiler tagged 2A, October 20, 2026: the controller program was backed up, parameters reset/restored, and the uncleared interlock alarms subsequently cleared. This observed sequence does not establish that similar alarms are always caused by programming.

## 5. Preliminary Fault-Code / Symptom Register

### Provisional records

- B&G Ecocirc — E02: field-reported high motor current. Investigate mechanical, motor, and electrical causes using the exact model's procedure.
- B&G Ecocirc — E06: field-reported incoming voltage imbalance. Investigate supply and connection conditions using qualified electrical procedures.
- Tekmar 671 — outdoor sensor open-circuit alarm: verify the selected outdoor-source configuration, network/time settings, and validity of the configured data source.
- AERCO BME — external interlock alarm: verify the interlock circuit, BMS interface, and configuration.
- Danfoss VLT HVAC Drive — "FA04" question: confirm the exact drive model and whether FA04 is a drive parameter, displayed code, or external-document reference before diagnosing.

### Source and verification rules

- These are provisional field notes, not validated fault-code database entries. Do not invent source page numbers or treat observations as official manufacturer definitions.
- Manufacturer fault meanings, causes, checks, limits, and procedures must come from the exact applicable manual and cite the manual document and page.
- Keep field observations separate from verified manufacturer guidance. Include model, code/symptom, meaning, causes, checks, source document, and source page where supported.

## 6. Safety and Use Limits

### Use

- This tool provides suggestions only. Follow manufacturer procedures, site safety requirements, and qualified-person requirements.

- Apply lockout/tagout before work requiring equipment isolation. Follow electrical safety and refrigerant-handling requirements where applicable.
- Never bypass a safety device, interlock, or protective control to force equipment operation.
- The outdoor-reset calculator is a linear approximation; use the controller manual's curve for exact values.
- Enter target delta-T, superheat, and subcooling ranges from the data plate or manual; do not assume universal target ranges.
- Generic decision trees are standard-practice drafts and should be reviewed by a qualified person. Image-only scanned PDF pages may not be searchable.

# 7. Recommended Next Build Steps

## Project integration checklist

- 1. Add and organize manuals under pdfs/<manufacturer>/ (for example, pdfs/tekmar/).
- 2. Run npm run build, then npm run find-fault-pages to locate likely fault-code tables.
- 3. Transcribe verified fault codes into data/fault-codes/<manufacturer>.json with exact source document and page.
- 4. Create decision trees in data/trees/ using the project's actual schema and matching each filename to its tree ID.
- 5. Run npm run validate and npm test; resolve all citation and schema errors.
- 6. Generate refrigerant pressure-temperature tables with CoolProp only after checking the project script's requirements. Verify computed R-454B data against a published manufacturer chart before field use.
- 7. Test all four tabs in Chrome and Edge, including with the network disconnected.
- 8. Preserve package-lock.json without private registry URLs.

## Priority order

- First: Tekmar controls, B&G Ecocirc faults, and AERCO BME interlocks because prior cases provide useful field observations.
- Next: Danfoss VFD commissioning/programming, then additional equipment families as manuals and documented cases become available.
- To create drop-in JSON files, use the project's existing schema and validator; the decision-tree schema was not included in the supplied description.